Wokwi files:

sketch.ino file:

```
/*
  ==========================================================
      Dual Role IoT Monitoring System
      ESP32 + MQTT + Wokwi
  ==========================================================
*/

#include <WiFi.h>
#include <PubSubClient.h>
#include <DHT.h>
#include <ArduinoJson.h>

// ---------- WiFi ----------
const char* ssid = "Wokwi-GUEST";
const char* password = "";

// ---------- MQTT ----------
const char* mqttServer = "broker.hivemq.com";
const int mqttPort = 1883;

WiFiClient espClient;
PubSubClient client(espClient);

// ---------- Pins ----------
#define DHTPIN 15
#define DHTTYPE DHT22

#define HEART_PIN 34
#define SPO2_PIN 35
#define OXYGEN_PIN 32
#define AQI_PIN 33

DHT dht(DHTPIN, DHTTYPE);

// ---------- Sensor Values ----------

float roomTemp = 25;

float bodyTemp = 36.8;

int heartRate = 75;
```

```
int spo2 = 98;

int oxygen = 21;

int aqi = 40;

// ---------- Timing ----------

unsigned long previousMillis = 0;

const long interval = 5000;

// ---------- MQTT Topics ----------

const char* medicalTopic =
"hospital/medical";

const char* facilityTopic =
"hospital/facility";

const char* alertTopic =
"hospital/alerts";

void connectWiFi(){

  Serial.println();

  Serial.println("Connecting WiFi...");

  WiFi.begin(ssid,password);

  while(WiFi.status()!=WL_CONNECTED){

      delay(500);

      Serial.print(".");

  }

  Serial.println();

  Serial.println("WiFi Connected");
```

```
}

void connectMQTT(){

    while(!client.connected()){

        Serial.println("Connecting MQTT...");

        String id="ESP32-";

        id+=String(random(10000));

        if(client.connect(id.c_str())){

            Serial.println("MQTT Connected");

        }

        else{

            Serial.print("Retry...");

            delay(2000);

        }

    }

}

void readSensors() {

  roomTemp = dht.readTemperature();

  if (isnan(roomTemp))
    roomTemp = 25.0;

  heartRate = map(
      analogRead(34),
      0,
      4095,
      55,
      130
  );
```

```
  spo2 = map(
    analogRead(35),
    0,
    4095,
    85,
    100
  );

  oxygen = map(
    analogRead(32),
    0,
    4095,
    15,
    23
  );

  aqi = map(
    analogRead(33),
    0,
    4095,
    0,
    200
  );

  bodyTemp = 36.5 + random(-5,6)/10.0;

}

void publishData(){

StaticJsonDocument<256> medical;

medical["heartRate"]=heartRate;

medical["spo2"]=spo2;

medical["bodyTemp"]=bodyTemp;

char med[256];

serializeJson(medical,med);

client.publish(medicalTopic,med);
```

```
StaticJsonDocument<256> facility;

facility["roomTemp"]=roomTemp;

facility["oxygen"]=oxygen;

facility["aqi"]=aqi;

char fac[256];

serializeJson(facility,fac);

client.publish(facilityTopic,fac);

}

void publishAlerts(){

if(heartRate<60)
client.publish(alertTopic,"Low Heart Rate");

if(heartRate>100)
client.publish(alertTopic,"High Heart Rate");

if(spo2<94)
client.publish(alertTopic,"Low SpO2");

if(bodyTemp>37.5)
client.publish(alertTopic,"High Body Temperature");

if(roomTemp>30)
client.publish(alertTopic,"High Room Temperature");

if(oxygen<19)
client.publish(alertTopic,"Low Oxygen Level");

if(aqi>120)
client.publish(alertTopic,"Poor Air Quality");

}
```

```
void setup() {

  Serial.begin(115200);

  delay(1000);

  Serial.println("Hospital IoT System");

  dht.begin();

  connectWiFi();

  client.setServer(mqttServer, mqttPort);

}


void loop(){

if(!client.connected())
connectMQTT();

client.loop();

unsigned long now=millis();

if(now-previousMillis>interval){

previousMillis=now;

readSensors();

publishData();

publishAlerts();

Serial.println();
Serial.println("============================");
Serial.println("Hospital IoT Monitoring");
Serial.println("============================");

Serial.print("Heart Rate : ");
```

```
Serial.print(heartRate);
Serial.println(" BPM");

Serial.print("SpO2 : ");
Serial.print(spo2);
Serial.println("%");

Serial.print("Body Temp : ");
Serial.print(bodyTemp);
Serial.println(" C");

Serial.print("Room Temp : ");
Serial.print(roomTemp);
Serial.println(" C");

Serial.print("Oxygen : ");
Serial.print(oxygen);
Serial.println("%");

Serial.print("AQI : ");
Serial.println(aqi);

Serial.println("============================");

Serial.print("Heart Rate : ");
Serial.println(heartRate);

Serial.print("SpO2 : ");
Serial.println(spo2);

Serial.print("Body Temp : ");
Serial.println(bodyTemp);

Serial.print("Room Temp : ");
Serial.println(roomTemp);

Serial.print("Oxygen : ");
Serial.println(oxygen);

Serial.print("AQI : ");
Serial.println(aqi);

}
```

}

diagram.json file:


```
{
  "version": 1,
  "author": "Karthick",
  "editor": "wokwi",
  "parts": [
    { "type": "wokwi-esp32-devkit-v1", "id": "esp", "top": 0, "left": 0, "attrs": {} },
    { "type": "wokwi-dht22", "id": "dht1", "top": -182.1, "left": 244.2, "attrs": {} },
    { "type": "wokwi-potentiometer", "id": "pot1", "top": -240, "left": -260, "attrs": {} },
    { "type": "wokwi-potentiometer", "id": "pot2", "top": -80, "left": -260, "attrs": {} },
    { "type": "wokwi-potentiometer", "id": "pot3", "top": 75.5, "left": -259.4, "attrs": {} },
    { "type": "wokwi-potentiometer", "id": "pot4", "top": 240, "left": -260, "attrs": {} }
  ],
  "connections": [
    [ "esp:3V3", "dht1:VCC", "red", [] ],
    [ "esp:GND", "dht1:GND", "black", [] ],
    [ "esp:15", "dht1:SDA", "green", [] ],
    [ "esp:3V3", "pot1:VCC", "red", [] ],
    [ "esp:GND", "pot1:GND", "black", [] ],
    [ "esp:34", "pot1:SIG", "green", [] ],
    [ "esp:3V3", "pot2:VCC", "red", [] ],
    [ "esp:GND", "pot2:GND", "black", [] ],
    [ "esp:35", "pot2:SIG", "green", [] ],
    [ "esp:3V3", "pot3:VCC", "red", [] ],
    [ "esp:GND", "pot3:GND", "black", [] ],
    [ "esp:32", "pot3:SIG", "green", [] ],
    [ "esp:3V3", "pot4:VCC", "red", [] ],
    [ "esp:GND", "pot4:GND", "black", [] ],
    [ "esp:33", "pot4:SIG", "green", [] ],
    [ "dht1:GND", "esp:GND.1", "black", [ "v0" ] ],
    [ "dht1:SDA", "esp:D15", "green", [ "v163.2", "h-86.3", "v38.4", "h-76.8" ] ],
    [ "pot1:GND", "esp:GND.2", "black", [ "v46.7", "h183", "v259.2" ] ],
    [ "pot2:GND", "esp:GND.2", "black", [ "v78.7", "h163.8", "v67.2" ] ],
    [ "pot3:GND", "esp:GND.2", "black", [ "v33.9", "h183", "v-28.8" ] ],
    [ "pot4:GND", "esp:GND.2", "black", [ "v37.1", "h144.6", "v-201.6" ] ],
    [ "pot1:SIG", "esp:D34", "green", [ "v75.5", "h201.8", "v144" ] ],
    [ "pot2:SIG", "esp:D35", "green", [ "v0" ] ],
    [ "pot3:SIG", "esp:D32", "green", [ "v28.8", "h201.2", "v-96" ] ],
    [ "pot4:SIG", "esp:D33", "green", [ "v17.9", "h144.2", "v-230.4" ] ]
  ],
  "dependencies": {}
}
```

libraries.txt file:
PubSubClient
ArduinoJson
DHT sensor library